

AI-Powered ECG Anomaly Detection

PROJECT SYNOPSIS OF PROJECT-I

BACHELOR OF TECHNOLOGY

Computer Science & Engineering

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Problem Statement

Artificial Intelligence is increasingly used in automated healthcare systems to analyze electrocardiogram (ECG) signals and detect cardiovascular anomalies, such as arrhythmias. However, advanced deep learning models trained on historical ECG data often function as opaque "black boxes," making it difficult for clinicians to interpret their decision-making processes. Furthermore, these models frequently suffer from poor generalization across diverse populations due to severe class imbalance, signal noise, and flawed evaluation methods that fail to strictly separate patient data (ignoring the inter-patient paradigm). This can lead to unreliable diagnostic predictions where models might evaluate heartbeats based on memorized patient-specific artifacts or background noise rather than genuine electrophysiological markers, thereby reducing clinical trust and patient safety. Additionally, the high computational complexity of these models limits their real-time deployment on resource-constrained wearable devices. The objective of this project is to develop an explainable and hardware-efficient AI-based ECG anomaly detection system using hybrid Deep Learning techniques (such as CNNs, Bi-LSTMs, and Transformers) and Explainable AI (such as SHAP) to ensure accurate, transparent, and unbiased cardiovascular evaluation for continuous remote monitoring.

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Expected outcome

The expected outcome of this project is to address the "black-box" limitations and evaluation biases in AI-based electrocardiogram (ECG) anomaly detection systems and evaluate methods to mitigate these issues. The project will develop advanced hybrid deep learning models (such as CNN-Bi-LSTM-Transformers or LSTM autoencoders) to analyze raw ECG data and measure true diagnostic generalizability using strict inter-patient evaluation protocols.

The study will evaluate the performance and transparency of the models by applying Explainable AI techniques, specifically SHAP (SHapley Additive exPlanations), to attribute diagnostic predictions to physiologically relevant ECG waveform segments. Furthermore, the results will provide insights into how algorithmic opacity and hardware constraints affect automated cardiovascular diagnostics, and how these barriers can be reduced using techniques from Explainable AI and model optimization (such as quantization and pruning) for wearable edge devices.

The final outcome will include experimental results (detailing accuracy, memory footprint, and inference latency), clinical interpretability evaluations, and a comprehensive research report that can contribute to future research in trustworthy, deployable, and explainable AI-powered cardiac monitoring systems.

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Abstract

Cardiovascular diseases remain the leading global cause of mortality, making continuous and accurate electrocardiogram (ECG) monitoring essential for early diagnosis. Traditional manual interpretation is time-consuming, while conventional machine learning methods rely heavily on subjective, handcrafted feature extraction that struggles to generalize to unseen data. Although deep learning architectures have improved automated anomaly detection, they frequently suffer from high computational demands, poor generalization across diverse patients, and a "black-box" nature that limits clinical trust.

Recent advancements propose **hybrid end-to-end deep learning models**—such as those combining Convolutional Neural Networks (CNNs), Bidirectional Long Short-Term Memory (Bi-LSTMs), and Transformer attention mechanisms—to simultaneously capture local morphological waveforms and long-range temporal dependencies directly from raw signals. To bridge the interpretability gap, state-of-the-art frameworks integrate **Explainable AI (XAI) techniques like SHapley Additive exPlanations (SHAP)** and attention maps, which attribute predictions to specific, clinically relevant ECG segments (e.g., QRS complexes or P-waves) so that algorithmic decisions align with cardiological domain knowledge.

Furthermore, deploying these complex models in real-world wearable monitors and Internet of Medical Things (IoMT) devices requires rigorous hardware optimization through techniques like **quantization and pruning** to achieve real-time latency and minimal memory footprints. Ultimately, combining robust spatio-temporal feature learning, clinical explainability, and edge-device feasibility paves the way for highly accurate, transparent, and continuous automated cardiac monitoring systems.

Motivation

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally, making continuous and accurate electrocardiogram (ECG) monitoring essential for early diagnosis and treatment. Many healthcare systems now use artificial intelligence (AI) and deep learning-based tools to automate ECG analysis, helping to process large volumes of heartbeat data, reduce the time-consuming manual effort required by cardiologists, and mitigate inter-observer variability. However, the growing reliance on these automated diagnostic models has raised significant concerns regarding their clinical transparency, generalizability, and real-world deployment feasibility.

One major challenge associated with AI-driven ECG systems is their "black-box" nature and susceptibility to biased evaluations. Deep learning models—such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks—automatically extract complex morphological and temporal features from raw signals, which can result in highly accurate predictions but severely lack interpretability. Furthermore, if models are trained and tested using intra-patient data (where beats from the same patient appear in both sets), the AI system may unintentionally learn and reproduce patient-specific traits rather than actual disease patterns. As a result, the models often fail to generalize when applied to completely unseen patients in clinical settings.

For example, complex deep learning architectures have historically been computationally expensive and memory-intensive. While they achieve near-expert performance in detecting arrhythmias like Atrial Fibrillation (AF) or Premature Ventricular Contractions (PVC), they demand substantial resources that are incompatible with low-power wearable edge devices, such as smartwatches or Holter monitors. Additionally, without Explainable AI (XAI) tools, a clinician cannot verify whether the system is making predictions based on critical physiological markers—like a widened QRS complex or an absent P-wave—or if it is simply reacting to background noise.

The motivation for this project is to address the problems of clinical opacity and hardware constraints in automated ECG anomaly detection systems, promoting fairness and reliability in AI-driven cardiovascular diagnostics. By applying techniques from Deep Learning, Explainable AI (such as SHAP—SHapley Additive exPlanations), and model optimization (like quantization and pruning), this research aims to analyze how arrhythmias are detected

and explore methods to make these systems transparent and hardware-efficient. The project seeks to contribute to the development of responsible AI systems that ensure accurate and interpretable evaluation of cardiac conditions, regardless of inter-patient variability.

Ensuring explainability and edge-device feasibility in AI-based ECG systems is essential for building clinical trust and promoting widespread, continuous remote monitoring. The findings of this study may help healthcare organizations understand the risks of black-box algorithms and encourage the adoption of robust, ethical, and real-time AI practices in cardiovascular care.

Literature Review

Artificial Intelligence is widely used in modern healthcare systems to automatically analyze electrocardiogram (ECG) signals and detect cardiac anomalies. Many clinical settings use deep learning-based diagnostic tools to reduce the time-consuming manual effort required by cardiologists and to improve continuous monitoring efficiency. However, several studies have shown that these automated systems face significant challenges, such as the **"black-box" nature of deep learning, high computational demands, and poor generalization across different patients**, which can limit their reliability and clinical trust in real-world deployment. Recent research has focused on improving the interpretability, spatial-temporal modeling, and deployment feasibility of AI-based ECG systems. For example, **Manal Alghieth (2025)** proposed DeepECG-Net, integrating CNNs and transformer architectures to efficiently learn both local and global ECG signal variations while minimizing computational overhead and maintaining real-time inference. Similarly, **Mohammed T. Al-Bairmani et al. (2026)** proposed an explainable, multi-label arrhythmia detection system using a joint CNN-Bi-LSTM-transformer model. To improve clinical transparency, their research integrated **SHAP (SHapley Additive exPlanations)**, which attributes model predictions to physiologically relevant waveform segments, maintaining high accuracy while ensuring human-level interpretability.

Another study by **K. Ammar et al. (2025)** conducted experiments using an end-to-end deep learning framework that combines CNNs, Bidirectional LSTMs, and a multi-head attention mechanism. Their research demonstrated that models can effectively focus on critical segments of raw ECG signals without relying on biased manual feature extraction, significantly outperforming traditional methods in detecting underrepresented arrhythmias.

Additionally, **K. Nayana and S. Vinay (2026)** used unsupervised machine learning algorithms, specifically LSTM autoencoders, to analyze reconstruction errors in normal cardiac rhythms.

To address hardware and deployment constraints, **Mudassir Hasan Khan et al. (2025/2026)** analyzed the convergence of ECG foundation models and medical Large Language Models (LLMs) to create **"agentic cardiovascular intelligence"**. Their comprehensive survey showed that by applying compression techniques like quantization and pruning, these advanced, knowledge-aware AI models can be successfully deployed on resource-constrained wearable edge devices, providing real-time, privacy-preserving diagnostics.

Table 1: Literature Review of AI-Powered ECG Anomaly Detection

Title	Author(s)	Year	Methodology	Model	Result (Pros)	Limitations (Cons)
DeepECG-Net: a hybrid transformer-based deep learning model for real-time ECG anomaly detection	Manal Alghieth	2025	Utilizes a hybrid architecture combining CNNs for local feature extraction and transformers for global dependencies, along with an adaptive anomaly detection module and federated learning framework.	DeepECG-Net (Transformer-CNN Hybrid)	Achieves 98.2% accuracy, 96.8% recall, and 97.5% F1 score. Features low memory usage (30 MB) and sub-50 ms latency, making it ideal for wearable standalone platforms while preserving privacy.	The paper does not explicitly mention the specific type of Transformer being used in its methodology section. Like many hybrid models, it may carry higher computational complexity during training, though inference is optimized.
A joint CNN-Bi-LSTM-transformer architecture with SHAP explanations for multi-label arrhythmia detection from 12-lead ECGs	Mohammed T. Al-Bairmani, Mohammadreza Yazdchi, Fahimeh Nasimi	2026	Jointly extracts morphological, temporal, and spatial patterns from multi-label 12-lead ECGs. Integrates SHAP for case-by-case explainability.	CNN-Bi-LSTM-Transformer with SHAP	Achieves 91.24% accuracy and 83.01% macro-F1 score across 27 arrhythmia classes. Highly efficient with < 2 MB memory footprint and < 10 ms inference latency, providing human-level interpretability.	Validation is primarily on a single multi-institutional dataset (PhysioNet 2020); future work must address cross-institutional domain adaptation and real-time motion artifact suppression.
ECG Signal Analysis and Abnormality Detection Application	A. Jandera, Y. Petryk, M. Muzelak, T. Skovranek	2025	Processes raw ECG signals using standard methods (Pan-Tompkins, time windowing) combined with machine learning for multi-class, multi-label diagnostic categorization.	Support Vector Machine (SVM)	Effective on small datasets and features an interpretable kernel space with fast computational speed.	Single-lead ECG constraints limit spatial diagnostic information compared to 12-lead setups. Generalizability across diverse populations is limited due to single-dataset validation and susceptibility to noise.
ECG-CBA: An End-to-End Deep Learning Model for ECG Anomaly Detection Using CNN, BiLSTM, and Attention Mechanism	K. Ammar, S. Fraihat, G. Al-Naymat, Y. Sanjalawe	2025	An end-to-end framework learning directly from raw signals without handcrafted features. Uses CNN for spatial patterns, Bi-LSTM for temporal dependencies, and a multi-head attention mechanism.	ECG-CBA (CNN + Bi-LSTM + Attention)	Demonstrates 99.60% accuracy on the ECG5000 dataset and 98.80% on MIT-BIH. Robust against noise and inter-patient variability.	Ablation studies show noticeable performance drops in recall and accuracy if the Attention or Bi-LSTM layers are removed. Focuses heavily on binary classification (normal

						vs. abnormal).
Adaptive Toeplitz convolution-enhanced classifier for anomaly detection in ECG big data	Lili Wu, Tao Li, Majid Khan Majahar Ali, Chenmin Ni, Ying Tian, Xiaojie Zhou	2025	Employs multi-stage filtering and denoising (FDL) alongside an adaptive Toeplitz matrix to replace standard autoencoder encoding, feeding into traditional machine learning classifiers.	ECG-ATC combined with classifiers (KNN, DT, RF, XGBoost, etc.)	Achieves over 99.6% accuracy. Reduces computational complexity and runtime drastically while improving F1-scores and generalization.	Toeplitz convolution introduces nonlinear features making it unsuitable for linear classifiers like Logistic Regression and SVMs. Requires careful tuning of preprocessing stage parameters.
Deep learning for ECG-based cardiovascular disease diagnosis: a systematic review and meta-analysis	Abdullah Al-Mamun Bulbul, Md. Abdul Awal, Niruthikha Mahendran, Sheikh Mohammed Shariful Islam, Ganna Pogrebna, Allen G. Ross, Mohammad Ali Moni	2026	Systematic review and meta-analysis of 85 articles following PRISMA guidelines, utilizing HSROC curves to assess state-of-the-art DL algorithms.	Systematic Review (Evaluating CNNs, RNNs, LSTMs, Hybrids)	Meta-analysis indicates excellent diagnostic capability with summary operating points at 96% specificity and 97% sensitivity across the literature.	Literature exhibits severe gaps: lack of external cross-validation (only 35.3% of studies), virtually no uncertainty quantification (0%), and very sparse use of explainable AI (9.4%). Review limited to English peer-reviewed journals.
Advances in machine and deep learning for ECG beat classification: a systematic review	Allam Jaya Prakash, Abdelkader Nasreddine Belkacem, Ibrahim M. Elfadel, Herbert F. Jelinek, Mohamed Atef	2025	Systematic review of 106 articles published between 2014 and 2024 (PRISMA criteria), analyzing pre-processing, feature engineering, and automated DL architectures.	Systematic Review (Evaluating SVM, RF, CNN, LSTM, Transformers)	Identifies CNN-LSTM hybrids and attention networks as the current state-of-the-art (98-99% accuracy). Emphasizes inter-patient evaluation as the best practice for realistic deployment.	Identifies major field limitations: severe data imbalance, inter-patient variability, overfitting, and the computationally demanding nature of deep models. Review excluded non-English and grey literature.
A Systematic Review of ECG Arrhythmia Classification: Adherence to Standards, Fair Evaluation, and Embedded Feasibility	Guilherme A. L. Silva, Pedro H. L. Silva, Gladston J. P. Moreira, Vander L. S. Freitas, Jadson C. Gertrudes, Eduardo J. S.	2024/2025 (Reviews 2017-2024)	Systematic review of 122 articles focusing on E3C criteria (Embedded, Clinical, and Comparative Criteria), assessing AAMI compliance and inter-patient paradigms.	Systematic Review	Highlights models that successfully balance accuracy with hardware limitations (e.g., energy consumption, model size) for realistic embedded deployment in	Found that only 4.1% of evaluated studies adhered to full E3C standards, showing widespread inflation of metrics via intra-patient testing. Searching was restricted to the SCI-EXPANDED index.

	Luz				wearable devices.	
ECG Foundation Models, Medical LLMs, and Edge AI for Cardiovascular Intelligence: A Survey (Title derived from text)	Mudassir Hasan Khan, Ahmad Nayfeh, Mudassir Masood, Ali Ahmad Al-Shaikhi, Muhammad Mahboob Ur Rahman, Tareq Y. Al-Naffouri	2025/2026 (Reviews 2020-2026)	Literature review analyzing the convergence of ECG foundation models (signal perception), medical LLMs (clinical reasoning), and edge-based deployment (quantization, SLMs).	Systematic Review / Survey	Proposes a unified roadmap for "agentic AI" cardiovascular intelligence. Highlights hybrid compression strategies achieving >100x size reductions for wearables.	Highlights significant privacy/security vulnerabilities (adversarial prompts, membership inference) at the edge/cloud layers. LLM reasoning requires extensive hardware constraints and suffers from hallucinations.
Explainable AI-Powered ECG Anomaly Detection Using SHAP and LSTM Autoencoders	K. Nayana, S. Vinay	2026	Uses an unsupervised learning approach where LSTM autoencoders learn to reconstruct normal rhythms. SHAP is used to explain the time-step contributions to reconstruction errors.	LSTM Autoencoder + SHAP	Yields 97.6% accuracy and 96.7% F1-score. Offers time-step-level transparency for clinical trust and uses a compact architecture suitable for real-time edge devices.	Requires mitigation of class imbalance inherent to ECG data. Though robust to noise, unsupervised autoencoders rely entirely on deviation from normal models, which can be sensitive if normal variants are highly heterogeneous.

Recent advancements in automated ECG anomaly detection have successfully transitioned from traditional manual feature extraction to sophisticated end-to-end deep learning architectures. State-of-the-art hybrid models, such as those combining CNNs, Bi-LSTMs, and Transformers, now excel at simultaneously capturing local waveform morphologies and long-range temporal dependencies. To overcome the "black-box" nature of these deep networks and build clinical trust, the integration of Explainable AI—particularly SHAP and attention mechanisms—provides transparent, physiologically relevant insights into diagnostic predictions. Additionally, researchers are actively optimizing these complex architectures through pruning and quantization to meet the strict computational and memory constraints of wearable edge devices. Ultimately, the convergence of robust spatio-temporal modeling, clinical explainability, and hardware efficiency is paving the way for highly reliable, real-time continuous cardiac monitoring systems.

Problem Formulation

Artificial Intelligence is increasingly being used in modern healthcare systems to automatically analyze electrocardiogram (ECG) signals and detect cardiac arrhythmias. These AI-based systems use deep learning techniques to extract complex spatial and temporal features from raw ECG data and predict whether a heartbeat is normal or anomalous. However, many studies have revealed that such automated systems face significant challenges regarding clinical interpretability, generalization, and deployment feasibility. A major issue is the "black-box" nature of deep neural networks, where the model's decision-making process is entirely hidden from healthcare providers. For example, a highly complex model trained on imbalanced data might achieve excellent accuracy on paper, but may actually be relying on background noise or memorized patient-specific traits rather than genuine electrophysiological markers (such as a widened QRS complex or an absent P-wave). If the training data does not strictly separate patients (the inter-patient paradigm), the AI model may learn to prefer these patient-specific artifacts, leading to models that completely fail to generalize to new, unseen patients. This opacity, combined with the massive computational requirements of standard deep learning models, prevents clinical trust and hinders the deployment of these systems on real-world, resource-constrained wearable devices.

Therefore, there is a need to develop a system that can accurately detect cardiac anomalies while providing transparent, physiologically relevant explanations for its predictions. The proposed project aims to analyze raw ECG datasets, develop a hybrid deep learning architecture, and implement Explainable AI (XAI) alongside hardware optimization techniques to ensure a clinically trustworthy, unbiased, and deployable diagnostic tool.

Objectives of the Project

The main objectives of this project are:

1. **To study** existing deep learning-based ECG classification systems and analyze their limitations regarding the "black-box" problem, data imbalance, and hardware constraints for edge deployment.
2. **To collect and preprocess** raw ECG datasets by applying systematic filtering, noise reduction, and segmentation techniques to prepare the time-series signals for optimal deep learning.
3. **To develop** a robust hybrid deep learning model (such as a combination of CNNs, Bi-LSTMs) that can automatically and efficiently classify multi-label cardiac arrhythmias directly from raw signals.
4. **To evaluate** the model's diagnostic performance and generalizability using strict fairness and validation protocols, specifically the inter-patient data partitioning paradigm, to prevent biased evaluations.
5. **To explore and apply** model optimization strategies (such as quantization and pruning) to significantly reduce the model's computational complexity, memory footprint, and inference latency.
6. **To develop** a highly accurate, transparent, and hardware-efficient automated ECG anomaly detection framework that promotes clinical trust and supports continuous, remote cardiovascular monitoring on wearable edge devices.

Methodology / Planning of Work

The proposed project aims to develop a robust, explainable, and hardware-efficient system for detecting cardiac anomalies in electrocardiogram (ECG) signals using advanced deep learning architectures. The methodology involves several stages, including data collection, preprocessing, spatiotemporal feature learning, model development, interpretability integration, and hardware optimization.

1. Data Collection: The first step involves collecting standardized ECG datasets containing normal cardiac rhythms and various anomalies or arrhythmias (e.g., Atrial Fibrillation, Premature Ventricular Contractions). Publicly available, expert-annotated databases such as the MIT-BIH Arrhythmia Database, the PhysioNet/Computing in Cardiology Challenge datasets, or the ECG5000 dataset will be utilized for training and validation purposes.

2. Data Pre-processing: Raw ECG signals frequently suffer from physiological and environmental noises, requiring a multi-stage preprocessing pipeline. The data will first be resampled to a uniform frequency to ensure consistent temporal resolution. Digital filtering—such as high-pass filters to remove baseline wander, notch filters to eliminate powerline interference, and median or low-pass filters to suppress high-frequency muscle artifacts—will be applied. Finally, the signals will undergo amplitude normalization (e.g., Min-Max scaling to a or $[-1,1]$ range) to ensure consistent feature scaling and numerical stability during model training.

3. Segmentation and Feature Extraction: Continuous ECG waveforms will be segmented into fixed-length overlapping sliding windows (e.g., 4-second intervals or 140 time-steps) to capture transient arrhythmic events while optimizing computational memory. Instead of relying on manual, handcrafted feature extraction—which can introduce bias and fail to generalize—deep learning layers, specifically 1D-Convolutional Neural Networks (CNNs), will be employed to automatically extract localized morphological and spatial features (such as P-wave and QRS complex structures) directly from the raw signal.

4. Model Development : Hybrid deep learning architectures will be developed to classify the ECG beats and detect anomalies. The model will utilize CNNs for spatial feature extraction, coupled with Bidirectional Long Short-Term Memory (Bi-LSTM) networks to process sequences both forwards and backwards, effectively capturing long-range temporal dependencies and rhythm transitions. Additionally, a Multi-Head Attention mechanism or Transformer encoder will be integrated to dynamically weight and focus on the most diagnostically critical ECG segments. Alternatively, unsupervised LSTM autoencoders can be

trained to learn normal rhythms and detect anomalies via reconstruction error analysis.

5. Model Interpretability and Explainability (XAI): To overcome the "black-box" nature of deep neural networks and build clinical trust, Explainable AI (XAI) frameworks will be integrated. SHapley Additive exPlanations (SHAP) will be applied to the model's predictions to provide case-by-case transparency. This step generates visual attributions (such as saliency maps) that map anomaly predictions to specific, physiologically relevant time steps or waveform segments, ensuring the model's reasoning aligns with established cardiological guidelines.

6. Hardware Optimization: for Edge Deployment To ensure the system can be practically deployed in continuous, real-time wearable monitors or embedded devices, model compression techniques will be applied. Methods such as weight quantization (e.g., reducing 32-bit floating-point parameters to 8-bit integers) and network pruning will be used to significantly decrease the memory footprint, computational complexity, and energy consumption of the neural networks, facilitating ultra-low latency inference on edge devices.

7. Model Evaluation: The performance of the model will be rigorously evaluated using standard classification metrics, including overall accuracy, precision, recall, macro-F1 score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Crucially, to avoid biased evaluations and artificial metric inflation, the testing will follow the inter-patient paradigm, ensuring that the training and testing sets are strictly separated by patient to verify the model's true generalizability to unseen individuals.

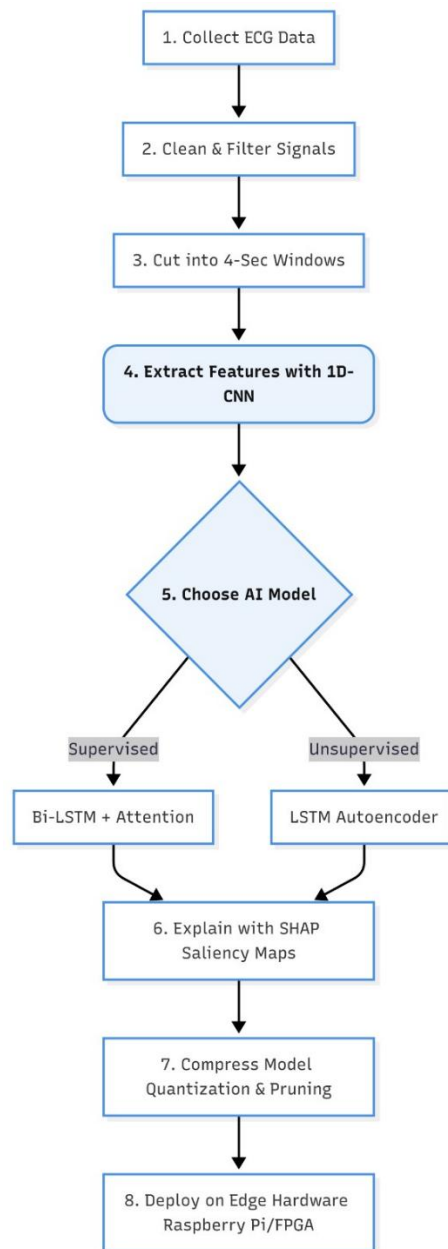


Figure 1: Architecture of the AI-Powered ECG Anomaly Detection

8. Result Analysis and Implementation: Finally, the results will be analyzed to assess the system's effectiveness in balancing diagnostic accuracy, clinical explainability, and hardware efficiency. The optimized model may be deployed on a prototype embedded platform (e.g., a Raspberry Pi or FPGA) to demonstrate real-time, privacy-preserving ECG anomaly screening in a simulated wearable environment.

Facilities Required for Proposed Work

Hardware Requirements

1. **Training Environment:** High-performance Laptop/Desktop or cloud instance with a minimum of 16 GB RAM and an Intel Core i3/i5 processor (or equivalent).
2. **Graphics Processing Unit (GPU):** An NVIDIA GPU (e.g., RTX 3060, Tesla T4, or A100) is highly recommended for accelerating the training of complex hybrid deep learning models (CNNs, Bi-LSTMs, Transformers).
3. **Storage:** At least 256 GB to 512 GB of storage to handle large, high-resolution multi-lead ECG databases.
4. **Edge Deployment Hardware (Optional/Testing):** Resource-constrained platforms such as a Raspberry Pi 4B (4 GB RAM) or an FPGA (e.g., Artix-7, ZYNQ Ultrascale) to validate real-time inference, latency, and energy consumption in wearable environments.
5. Stable internet connection for accessing cloud-based platforms (like Google Colab) and downloading large physiological datasets.

Software Requirements

1. **Programming Language:** Python 3.8+ (and optionally MATLAB for digital signal processing GUI application design).
2. **Development Platforms:** Google Colab, Kaggle, or local Jupyter environments.

Libraries and Frameworks:

1. TensorFlow / Keras / PyTorch: For building, training, and optimizing deep neural network architectures.
2. SciPy: Specifically its signal module, for vital pre-processing tasks like high-pass/low-pass digital filtering and noise reduction.
3. Scikit-learn: For evaluating classical machine learning classifiers, model metrics, and data splitting.
4. SHAP (SHapley Additive exPlanations): Specifically tools like *DeepExplainer* to visualize feature importance and provide case-by-case clinical interpretability.
5. NumPy & Pandas: For data manipulation and matrix operations.

3. Visualization Tools:

1. Matplotlib

2. Seaborn

Dataset Requirements

Standardized, expert-annotated ECG datasets will be collected from public medical repositories such as PhysioNet or Kaggle. Primary databases for training and validation include the MIT-BIH Arrhythmia Database, the PhysioNet/Computing in Cardiology Challenge 2020 dataset, the PTB-XL dataset, and the ECG5000 dataset. These datasets ensure the model is trained on a wide variety of normal rhythms and multi-label anomalous heartbeats.

Documentation Tools

1. Microsoft Word for documentation and report writing.
2. Microsoft Excel for literature review tables, hyperparameter tracking, and data analysis.

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